

# Christoph Junghans

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Group Leader

Applied Computer Science Group

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## Research Interests

- **Currently:** Applied computer science
- Software design and management – core developer of the coarse-graining package [VOTCA](#)
- Porting and optimization of scientific codes ([VOTCA](#), [Cabana](#), [FleCSI](#))
- Novel software development workflow for Exascale computing
- Systematic coarse-graining of liquids and implementation of new methodologies (e.g. thermostats)
- Development of adaptive resolution simulation approaches (changing the degrees of freedom on the fly)
- Generalized ensemble methods to study phase transitions (multicanonical techniques)

## Education

- Oct 2010 **Dr. rer. nat. (Ph.D.) in Physics**, *magna cum laude*,  
Johannes Gutenberg University, Mainz, Germany  
[Between the Scales: Water from different Perspectives](#)  
An adaptive resolution molecular dynamics study using various coarse-graining methods
- Nov 2006 **Diploma (M.Sc.) in Physics**, *summa cum laude*, University of Leipzig, Germany  
[Aggregation of Mesoscopic Protein-like Heteropolymers](#)  
A generalized ensemble Monte Carlo study using multicanonical techniques

## Work Experience

- Feb 2021 - Present **Group Leader**, Applied Computer Science Group, Computer, Computational, and Statistical Sciences Division, Los Alamos National Lab
- Feb 2016 - Present **CNLS Affiliate**, Center for Nonlinear Studies, Theoretical Division, Los Alamos National Lab
- July 2018 - Feb 2021 **Deputy Group Leader**, Applied Computer Science Group, Computer, Computational, and Statistical Sciences Division, Los Alamos National Lab
- Jan 2017 - July 2018 **Deputy Team Leader**, Co-Design Team, Computer, Computational, and Statistical Sciences Division, Los Alamos National Lab
- May 2014 - July 2018 **Staff Scientist**, Computer, Computational, and Statistical Sciences Division, Los Alamos National Lab
- Mar 2013 – Apr 2014 **Director's Postdoc. Fellow**, Theoretical Division, Los Alamos National Lab
- Nov 2011 – Feb 2013 **Postdoc. Research Assistant**, Theoretical Division, Los Alamos National Lab
- May 2012 – June 2012 **Member**, The Kavli Institute for Theoretical Physics, University of California, Santa Barbara, CA
- Nov 2010 – Oct 2011 **Postdoc. Research Assistant**, Polymer Theory Group, Max Planck Institute for Polymer Research, Mainz, Germany
- Jan 2007 – Oct 2010 **Ph.D. Student**, Polymer Theory Group, Max Planck Institute for Polymer Research, Mainz, Germany
- Jan 2009 – July 2009 **Specialist for Application Performance & Deep Computing**, Internship, IBM Systems & Technology Group Europe
- Oct 2003 – Sept 2006 **Student Assistant**, Institute for Theoretical Physics, University of Leipzig

|                       |                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------|
| Aug 2005 – Oct 2005   | <b>Research Student</b> in the “Scientific Computing” program, Jülich Supercomputing Centre, Germany       |
| Sep 2004              | <b>Student Assistant</b> , Institute for Meteorology, University of Leipzig                                |
| Feb 2003 – April 2003 | <b>Research Student</b> , Chair for Fluid Mechanics, Martin Luther University of Halle-Wittenberg, Germany |

## Open Source Projects

|                           |                                                             |                      |
|---------------------------|-------------------------------------------------------------|----------------------|
| <a href="#">VOTCA</a>     | Versatile Toolkit for coarse-graining applications          | Core Developer       |
| <a href="#">Cabana</a>    | Performance-portable library for particle-based simulations | Core Developer       |
| <a href="#">FleCSI</a>    | Flexible Computer Science Infrastructure                    | Developer            |
| <a href="#">Fedora</a>    | Linux distribution                                          | Contributor          |
| <a href="#">OpenSuse</a>  | Linux distribution                                          | Contributor          |
| <a href="#">ESPReSo++</a> | Successor of the ESPResSo simulation package                | Developer            |
| <a href="#">GroMaCS</a>   | Versatile package to perform molecular dynamics             | Developer (inactive) |
| <a href="#">ESPReSo</a>   | Extensible Simulation Package for Research on Soft matter   | Developer (inactive) |

More information can be found on [my GitHub profile](#) and [my OpenHUB profile](#).

## Awards

|           |                                                                                          |
|-----------|------------------------------------------------------------------------------------------|
| Nov 2019  | Top 10% most cited PLOS ONE papers published in 2015                                     |
| Sep 2017  | LAAP Award in recognition of outstanding contributions to the ISTI Summer School Program |
| Feb 2017  | Postdoc. Distinguished Mentor Award, LANL                                                |
| July 2016 | LAAP Achievement Award in appreciation for active service on the WSST team               |
| Jan 2015  | Classified as outstanding researcher, USCIS                                              |
| Jan 2014  | Most cited article published in EPJE in the past 5 years                                 |
| Mar 2013  | Los Alamos National Laboratory Director’s Post-Doctoral Fellowship                       |
| May 2012  | Member of the Kavli Institute for Theoretical Physics                                    |
| June 2010 | Participant of the 60th Lindau Nobel Laureate Meeting                                    |
| Nov 2006  | Diploma with honor (highest possible grade)                                              |
| Nov 2005  | Wolfgang Natonek award, University of Leipzig                                            |
| Oct 2003  | Teubner award, Department for Physics and Earth Science, University of Leipzig           |
| July 2000 | Book award of the German Physical Society (DPG) for high school graduates                |

## Professional Skills

### *Managerial*

- Line management, since July 2018, Group of 65+ staff scientists and Postdocs, \$50M+ budget
- Team management, 2 years, technical management of 12+ team member
- Project management, multiple lead roles in Exascale Computing Project (ECP), Laboratory Directed Research & Development (LDRD) and Advanced Simulation & Computing (ASC) projects

### *Referee for Scientific Journals*

- ACS: J. Chem. Inf. Model. & J. Phys. Chem.
- AIP: J. Chem. Phys.
- Elsevier: Comp. Phys. Comm.
- RSC: Phys. Chem. Chem. Phys.
- and others

*Code Review*

- [Gromacs](#)
- [OpenSuse Build Service](#)

*Conferences*

- Member of the Organization Committee for the [Salishan Conference on High Speed Computing](#)
- Chairman for APS annual meetings

*Representative*

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|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nov 2018 - Sep 2022   | Member of the Information Science and Technology Institute (ISTI) council, LANL                                                                                                                                                                             |
| July 2015 - June 2016 | Member of the Worker Safety & Security Team (WSST) for CCS-7, LANL                                                                                                                                                                                          |
| July 2007 - Dec 2008  | PhD representative of the theory group, Max Planck Institute for Polymer Research, Mainz                                                                                                                                                                    |
| Nov 2007 - Nov 2008   | PhD representative of the MPI for Polymer Research                                                                                                                                                                                                          |
| Oct 2002 - Sep 2005   | Elected member of student government (Fachschaftratsrat) of the Department for Physics and Earth Science, University of Leipzig - Extensive committee work in this period incl. spokesperson, treasurer and member of several faculty/university committees |

**Computer Skills**

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|-------------------------|--------------------------------------------|
| <b>General</b>          | Linux, AIX, Mac OS, DOS, Windows           |
| <b>Programming</b>      | C, Fortran, MPI, OpenMP, C++               |
| <b>Scripting</b>        | Shell, awk, Perl, expect, PHP, tcl, Python |
| <b>Markup Languages</b> | HTML, latex, mediawiki, txt2tags           |

**Teaching Experience**

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|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aug 2021             | “VOTCA and LAMMPS for electronic spectroscopy and transport” (Tutorial), <a href="#">LAMMPS Workshop and Symposium</a> , Virtual                                                 |
| Aug 2019             | “VOTCA and LAMMPS for electronic spectroscopy and transport” (Tutorial), <a href="#">LAMMPS Workshop and Symposium</a> , Albuquerque, NM                                         |
| May 2018 - Aug 2018  | <a href="#">Co-Design Summer School</a> (6 Students), Los Alamos National Laboratory                                                                                             |
| May 2017 - Jul 2017  | <a href="#">Co-Design Summer School</a> (6 Students), Los Alamos National Laboratory                                                                                             |
| Oct 2016             | “Coarse-Graining with VOTCA” (Tutorial), <a href="#">CECAM Workshop “Multiscale Simulations of Soft Matter with Hands-On Tutorials on ESPResSo++ and VOTCA”</a> , Mainz, Germany |
| Jun 2016 - Aug 2016  | <a href="#">Co-Design Summer School</a> (7 Students), Los Alamos National Laboratory                                                                                             |
| May 2016             | Gitlab Hands-On course, ASC/NGC Program, LANL                                                                                                                                    |
| Oct 2015             | “Coarse-Graining with VOTCA” (Tutorial), <a href="#">CECAM Summer School “Simulating Soft and Active Matter with ESPResSo, ESPResSo++ and VOTCA”</a> , ICP Stuttgart             |
| Sep 2015             | Git Hands-On course, ASC/NGC Program, LANL                                                                                                                                       |
| Jun 2015 - Aug 2015  | <a href="#">Co-Design Summer School</a> (6 Students), Los Alamos National Laboratory                                                                                             |
| Nov 2014             | Git IC Tutorial, Institutional Computing, LANL                                                                                                                                   |
| Oct 2014             | Hands-on: VOTCA, <a href="#">CECAM “School on Multiscale Modeling and Use of Espresso++ and VOTCA”</a> , Mainz, Germany                                                          |
| Jun 2014 - Aug. 2014 | <a href="#">Co-Design Summer School</a> (6 Students), Los Alamos National Laboratory                                                                                             |
| Aug 2014             | Gromacs Tutorial, <a href="#">The Eighth q-bio Summer School</a> , Albuquerque, NM (LA-UR 14-26188)                                                                              |

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|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jun 2014            | Gromacs IC Tutorial, Institutional Computing, LANL (LA-UR 14-24814)                                                                                                  |
| Oct 2013            | “Systematic Coarse-Graining with VOTCA” (Tutorial), <a href="#">CECAM Summer School “Simulating Soft Matter with ESPResSo, ESPResSo++ and VOTCA”</a> , ICP Stuttgart |
| Jul 2013 - Aug 2013 | <a href="#">Co-Design Summer School</a> (6 Students), Los Alamos National Laboratory                                                                                 |
| Jul 2013            | Gromacs Tutorial, <a href="#">The Seventh q-bio Summer School</a> , Santa Fe, NM                                                                                     |
| Oct 2012            | “Systematic Coarse-Graining with VOTCA” (Tutorial), <a href="#">CECAM Workshop “Simulating Soft Matter with ESPResSo, ESPResSo++ and VOTCA”</a> , ICP Stuttgart      |
| Jul 2012            | Gromacs Tutorial, <a href="#">The Sixth q-bio Summer School</a> , Santa Fe, NM                                                                                       |
| Oct 2011            | Hands-on: VOTCA, <a href="#">CECAM Workshop “Coarse-grained Simulation of Biological Soft Matter Systems using ESPResSo”</a> , ICP Stuttgart                         |
| May 2010            | Votca Workshop, <a href="#">CSI Darmstadt</a>                                                                                                                        |
| Apr 2008 - Sep 2008 | Theoretical physics III, Johannes Gutenberg University of Mainz                                                                                                      |
| Oct 2007 - Mar 2008 | Theoretical physics II, Johannes Gutenberg University of Mainz                                                                                                       |
| Apr 2007 - Sep 2007 | Theoretical physics I, Johannes Gutenberg University of Mainz                                                                                                        |
| Apr 2006 - Sep 2006 | Introduction to computer simulations I, University of Leipzig                                                                                                        |
| Sep 2004 - Oct 2004 | Mathematical preparation course for first-year students, University of Leipzig                                                                                       |

## Mentoring

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|---------------------|-----------------------------------------------------------------------------------|
| Feb 2022 - present  | Postdoc. Research Assistant, Los Alamos National Laboratory                       |
| Jun 2019 - Mar 2021 | Postdoc. Research Assistant, Los Alamos National Laboratory                       |
| Feb 2018 - Apr 2020 | Postdoc. Research Assistant (now Staff Scientist), Los Alamos National Laboratory |
| May 2018 - Feb 2019 | Postdoc. Research Assistant (now Staff Scientist), Los Alamos National Laboratory |
| Feb 2017 - Jan 2019 | Postdoc. Research Assistant (now Staff Scientist), Los Alamos National Laboratory |
| Sep 2018            | Member of PhD committee, University of Reims Champagne-Ardenne                    |
| Jun 2018 - Aug 2018 | ECP Summer Student, Los Alamos National Laboratory                                |
| May 2018 - Aug 2018 | Co-Design Summer School (6 Students), Los Alamos National Laboratory              |
| May 2018 - Aug 2018 | ASC Summer Student, Los Alamos National Laboratory                                |
| Feb 2016 - May 2018 | Postdoc. Research Assistant (now Staff Scientist), Los Alamos National Laboratory |
| Dec 2016 - Apr 2018 | Postdoc. Research Assistant (now Staff Scientist), Los Alamos National Laboratory |
| Jun 2017 - Aug 2017 | High School Internship Student, Los Alamos National Laboratory                    |
| Apr 2017 - Jun 2017 | ASC Summer Student, Los Alamos National Laboratory                                |
| May 2017 - Jul 2017 | Data-Science at Scale Summer Student, Los Alamos National Laboratory              |
| May 2017 - Jul 2017 | Co-Design Summer School (6 Students), Los Alamos National Laboratory              |
| May 2015 - May 2017 | Metropolis Postdoc. Fellow (now Staff Scientist), Los Alamos National Laboratory  |
| Aug 2016 - Apr 2017 | Postdoc. Research Assistant (now Staff Scientist), Los Alamos National Laboratory |
| Jul 2016 - Sep 2016 | CNLS Summer Student, Los Alamos National Laboratory                               |
| Jun 2016 - Aug 2016 | Co-Design Summer School (7 Students), Los Alamos National Laboratory              |
| Jul 2015 - Sep 2015 | IC Summer Student, Los Alamos National Laboratory                                 |
| Jun 2015 - Aug 2015 | Co-Design Summer School (6 Students), Los Alamos National Laboratory              |
| Jul 2014 - Sep 2014 | CNLS Summer Student, Los Alamos National Laboratory                               |
| Jun 2014 - Aug 2014 | Co-Design Summer School (6 Students), Los Alamos National Laboratory              |
| Jul 2013 - Aug 2013 | Co-Design Summer School (6 Students), Los Alamos National Laboratory              |
| Jun 2010 - Aug 2010 | Summer Student, University of Mainz                                               |
| Oct 2005 - Mar 2006 | Simulation methods in generalized ensembles (2 Students), University of Leipzig   |

## Talks

Over 50 invited and contributed talks

### Selected Talks

- Sep. 2018 Lecture: “Co-Design at Los Alamos National Laboratory”, [University of Reims Champagne-Ardenne](#)
- Oct. 2016 Talk: “Recent developments in VOTCA”, [CECAM Workshop “Multiscale Simulations of Soft Matter with Hands-On Tutorials on ESPResSo++ and VOTCA”](#), Mainz, Germany
- May 2012 Seminar: “Particle-based multi-scale simulations using the Adaptive Resolution Scheme”, [IBM Research, Almaden](#)
- July 2011 Colloquium: “Locality Analysis via Adaptive Resolution Simulations”, Theoriekolloquium, [Faculty of Natural Sciences II, Martin Luther University Halle-Wittenberg](#)

## Publications

h-index: [23 \(Google Scholar\)](#), [19 \(ResearcherID\)](#)

### Reviewed Papers

45. S. T. Reeve, J.-L. Fattebert, S. DeWitt, D. Joy, P. Seleson, S. Slattery, A. Scheinberg, R. Halver, **C. Junghans**, C. F. A. Negre, M. E. Wall, Y. Zhang, A. M. Niklasson, D. Perez, S. M. Mniszewski, and J. Belak, *Co-design for Particle Applications at Exascale*, [Comp. Sci. Eng. PrePrints\(2024\)](#).
44. S. Karra, M. Mehana, N. Lubbers, Y. Chen, A. Diaw, J. E. Santos, A. Pachaliev, R. S. Pavel, J. R. Haack, M. McKerns, **C. Junghans**, Q. Kang, D. Livescu, T. C. Germann, and H. S. Viswanathan, *Predictive Scale-Bridging Simulations through Active Learning*, [Sci. Rep. \*\*13\*\*, 16262 \(2023\)](#).
43. S. Wijesinghe, **C. Junghans**, D. Perahia, and G. S. Grest, *Polydots, soft nanoparticles, at membrane interfaces*, [RSC Adv. \*\*13\*\*, 19227 \(2023\)](#).
42. R. Halver, **C. Junghans**, and G. Sutmann, *Using heterogeneous GPU nodes with a Cabana-based implementation of MPCD*, [Paral. Comp. \*\*117\*\*, 103033 \(2023\)](#).
41. J. Vance, Z.-H. Xu, N. Tretyakov, T. Stuehn, M. Rampp, S. Eibl, **C. Junghans**, and A. Brinkmann, *Code modernization strategies for short-range non-bonded molecular dynamics simulations*, [Comp. Phys. Comm. \*\*290\*\*, 108760 \(2023\)](#).
40. R. Halver, **C. Junghans**, and G. Sutmann, *Kokkos-Based Implementation of MPCD on Heterogeneous Nodes*, in: R. Wyrzykowski, J. Dongarra, E. Deelman, and K. Karczewski (Eds.), *Parallel Processing and Applied Mathematics. PPAM 2022. Lecture Notes in Computer Science*, [Vol. \*\*13827\*\* \(2023\)](#).
39. A. Pachaliev, R. S. Pavel, J. E. Santos, A. Diaw, N. Lubbers, M. Mehana, J. R. Haack, H. S. Viswanathan, D. Livescu, T. C. Germann, and **C. Junghans**, *GLUE Code: A framework handling communication and interfaces between scales*, [J. Open Source Softw. \*\*7\*\*, 4822 \(2022\)](#).
38. P. Grete, J. C. Dolence, J. M. Miller, J. Brown, B. Ryan, A. Gaspar, F. Glines, S. Swaminarayan, J. Lippuner, C. J. Solomon, G. Shipman, **C. Junghans**, D. Holladay, and J. M. Stone, *Parthenon - a performance portable block-structured adaptive mesh refinement framework*, [Int. J. High Perf. Comp. App. \*\*37\*\*, 465 \(2023\)](#).
37. S. Slattery, S. T. Reeve, **C. Junghans**, D. Lebrun-Grandie, R. Bird, G. Chen, S. Fogerty, Y. Qiu, S. Schulz, A. Scheinberg, A. Isner, K. Chong, S. Moore, T. Germann, J. Belak, and S. Mniszewski, *Cabana: A Performance Portable Library for Particle-Based Simulations*, [J. Open Source Softw. \*\*7\*\*, 4115 \(2022\)](#).

36. J. Bakosi, R. Bird, F. Gonzalez, **C. Junghans**, W. Li, H. Luo, A. Pandare, and J. Waltz, *Asynchronous distributed-memory task-parallel algorithm for compressible flows on unstructured 3D Eulerian grids*, *Adv. Eng. Softw.* **160**, 102962 (2021).
35. S. M. Mniszewski, J. Belak, J.-L. Fattebert, C. F. A. Negre, S. R. Slattery, A. A. Adedoyin, R. F. Bird, C.S. Chang, G. Chen, S. Ethier, S. Fogerty, S. Habib, **C. Junghans**, D. Lebrun-Grandie, J. Mohd-Yusof, S. G. Moore, D. Osei-Kuffuor, S. J. Plimpton, A. Pope, S. T. Reeve, L. Ricketson, A. Scheinberg, A. Y. Sharma, and M. E. Wall, *Enabling Particle Applications for Exascale Computing Platforms*, *Int. J. High Perf. Comp. App.* **35**, 572 (2021).
34. A. Diaw, K. Barros, J. Haack, **C. Junghans**, B. Keenan, Y. W. Li, D. Livescu, N. Lubbers, M. McKerns, R. S. Pavel, D. Rosenberger, I. Sagert, and T. C. Germann, *Multiscale simulation of plasma flows using active learning*, *Phys. Rev. E* **102**, 023310 (2020).
33. N. Lubbers, A. Agarwal, Y. Chen, S. Son, M. Mehana, Q. Kang, S. Karra, **C. Junghans**, T. C. Germann, and H. S. Viswanathan, *Modeling and scale-bridging using machine learning: nanoconfinement effects in porous media*, *Sci. Rep.* **10**, 13312 (2020).
32. G. Tirimbo, V. Sundaram, O. Caylak, W. Scharpach, J. Sijen, **C. Junghans**, J. Brown, F. Zapata Ruiz, N. Renaud, J. Wehner, and B. Baumeier, *Excited-State Electronic Structure of Molecules Using Many-Body Green's Functions: Quasiparticles and Electron-Hole Excitations with VOTCA-XTP*, *J. Chem. Phys.* **152**, 114103 (2020).
31. H. V. Guzman, N. Tretyakov, H. Kobayashi, A. C. Fogarty, K. Kreis, J. Krajniak, **C. Junghans**, K. Kremer, and T. Stuehn, *ESPResSo++ 2.0: Advanced methods for multiscale molecular simulation*, *Comp. Phys. Comm.* **238**, 66 (2019).
30. J. Wehner, L. Brombacher, J. Brown, **C. Junghans**, O. Caylak, Y. Khalak, P. Madhikar, G. Tirimbo, and B. Baumeier, *Electronic Excitations in Complex Molecular Environments: Many-Body Green's Functions Theory in VOTCA-XTP*, *J. Chem. Theo. Comp.* **14**, 6253 (2018).
29. C. Krekeler, A. Agarwal, **C. Junghans**, M. Praprotnik, and L. Delle Site, *Adaptive Resolution Molecular Dynamics Technique: Down to the Essential*, *J. Chem. Phys.* **149**, 024104 (2018).
28. H. V. Guzman, **C. Junghans**, K. Kremer, and T. Stuehn, *Scalable and fast heterogeneous molecular simulation with predictive parallelization schemes*, *Phys. Rev. E* **96**, 053311 (2017).
27. S. K. Gutierrez, K. Davis, D. Arnold, R. S. Baker, R. W. Robey, P. McCormick, D. Holladay, J. A. Dahl, J. Zerr, F. Weik, and **C. Junghans**, *Accommodating Thread-Level Heterogeneity in Coupled Parallel Applications*, in: *Proceedings of 2017 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, Orlando, FL, USA, 469 (2017).
26. **C. Junghans**, A. Agarwal, and L. Delle Site, *Computational efficiency and Amdahl's law for the adaptive resolution simulation technique*, *Comp. Phys. Comm.* **215**, 20 (2017).
25. T. E. de Oliveira, P. A. Netz, K. Kremer, **C. Junghans**, and D. Mukherji, *C-IBI: Targeting cumulative coordination within an iterative protocol to derive coarse-grained models of (multi-component) complex fluids*, *J. Chem. Phys.* **144**, 174106 (2016).
24. R. S. Pavel, A. L. McPherson, T. C. Germann, and **C. Junghans**, *Database Assisted Distribution to Improve Fault Tolerance for Multiphysics Applications*, in: *Proceedings of the 2nd International Workshop on Hardware-Software Co-Design for High Performance Computing 2015 (Co-HPC '15)*, ACM, New York, NY, USA, Article 4.



23. S. Y. Mashayak, M. N. Jochum, K. Koschke, N. R. Aluru, V. Rühle, and **C. Junghans**, *Relative entropy and optimization-driven coarse-graining methods in VOTCA*, [PLoS ONE \*\*10\*\*, e131754 \(2015\)](#). “Among the top 10% most cited PLoS ONE papers published in 2015.” (2019)
22. S. M. Mniszewski, **C. Junghans**, A. F. Voter, D. Perez, and S. J. Eidenbenz, *TADSim: Discrete Event-based Performance Prediction for Temperature Accelerated Dynamics*, [Trans. Mod. Comp. Sim. \*\*25\*\*, 15 \(2015\)](#).
21. D. G. Roehm, R. S. Pavel, K. Barros, B. Rouet-Leduc, A. L. McPherson, T. C. Germann, and **C. Junghans**, *Distributed Database Kriging for Adaptive Sampling*, [Comp. Phys. Comm. \*\*192\*\*, 138 \(2015\)](#).
20. S. Bevc, **C. Junghans**, and M. Praprotnik, *STOCK: Structure Mapper and Online Coarse-Graining Kit for Molecular Simulations*, [J. Comp. Chem. \*\*36\*\*, 467 \(2015\)](#).
19. **C. Junghans**, D. Perez and T. Vogel, *Molecular Dynamics in the Multicanonical Ensemble: Equivalence of Wang-Landau Sampling, Statistical Temperature Molecular Dynamics, and Metadynamics*, [J. Chem. Theo. Comp. \*\*10\*\*, 1843 \(2014\)](#).
18. B. Rouet-Leduc, K. Barros, E. Cieren, V. Elango, **C. Junghans**, T. Lookman, J. Mohd-Yusof, R. S. Pavel, A. Y. Rivera, D. Roehm, A. L. McPherson, T. C. Germann, *Spatial adaptive sampling in multiscale simulation*, [Comp. Phys. Comm. \*\*185\*\*, 1857 \(2014\)](#).
17. A. Nagarajan, **C. Junghans** and S. Matysiak, *Multiscale simulation of liquid water using a four-to-one mapping for coarse-graining*, [J. Chem. Theo. Comp. \*\*9\*\*, 5168 \(2013\)](#).
16. S. Bevc, **C. Junghans**, K. Kremer and M. Praprotnik, *Adaptive resolution simulation of salt solutions*, [New J. Phys., \*\*15\*\*, 105007 \(2013\)](#).
15. P. Ganguly, D. Mukherji, **C. Junghans** and N. F. A. van der Vegt, *Kirkwood-Buff coarse-grained force fields for aqueous solutions*, [J. Chem. Theo. Comp., \*\*8\*\*, 1802 \(2012\)](#).
14. S. Fritsch, S. Poblete, **C. Junghans**, G. Ciccotti, L. Delle Site and K. Kremer, *Adaptive resolution molecular dynamics simulation through coupling to an internal particle reservoir*, [Phys. Rev. Lett. \*\*108\*\*, 170602 \(2012\)](#).
13. S. Fritsch, **C. Junghans** and K. Kremer, *Structure formation of toluene around C60: Implementation of the Adaptive Resolution Scheme (AdResS) into GROMACS*, [J. Chem. Theo. Comp. \*\*8\*\*, 398 \(2012\)](#).
12. V. Rühle and **C. Junghans**, *Hybrid approaches to coarse-graining using the VOTCA package: liquid hexane*, [Macromol. Theory Simul., \*\*20\*\*, 472 \(2011\)](#).
11. **C. Junghans**, W. Janke and M. Bachmann, *Hierarchies in Nucleation Transitions*, [Comp. Phys. Comm., \*\*182\*\*, 1937 \(2011\)](#).
10. B. P. Lambeth, Jr., **C. Junghans**, K. Kremer, C. Clementi, and L. Delle Site, *Communication: On the Locality of Hydrogen Bond Networks at Hydrophobic Interface*, [J. Chem. Phys. \*\*133\*\*, 221101 \(2010\)](#).
9. **C. Junghans** and S. Poblete, *A reference implementation of the adaptive resolution scheme in ESPResSo*, [Comp. Phys. Comm. \*\*181\*\*, 1449 \(2010\)](#).
8. V. Rühle, **C. Junghans**, A. Lukyanov, K. Kremer and D. Andrienko, *Versatile Object-oriented Toolkit for Coarse-graining Applications*, [J. Chem. Theo. Comp. \*\*5\*\*, 3211 \(2009\)](#).
7. **C. Junghans**, M. Bachmann and W. Janke, *Statistical Mechanics of Aggregation and Crystallization for Semiflexible Polymers*, [Europhys. Lett. \*\*87\*\*, 40002 \(2009\)](#).

6. H. Wang, **C. Junghans** and K. Kremer, *Comparative atomistic and coarse-grained study of water: What do we lose by coarse-graining?*, *Euro. Phys. J. E* **28**, 221 (2009). “Most cited article published in the journal in the past 5 years” (2014)
5. M. Praprotnik, **C. Junghans**, L. Delle Site and K. Kremer, *Simulation approaches to soft matter: Generic statistical properties vs. chemical details*, *Comp. Phys. Comm.* **179**, 51 (2008).
4. **C. Junghans**, M. Bachmann and W. Janke, *Thermodynamics of Peptide Aggregation Processes: An Analysis from Perspectives of Three Statistical Ensembles*, *J. Chem. Phys.* **128**, 085103 (2008) .
3. **C. Junghans**, M. Praprotnik and K. Kremer, *Transport properties controlled by a thermostat: An extended dissipative particle dynamics thermostat*, *Soft Matter* **4**, 156 (2008).
2. **C. Junghans**, M. Bachmann and W. Janke, *Microcanonical Analyses of Peptide Aggregation Processes*, *Phys. Rev. Lett.* **97**, 218103 (2006).
1. **C. Junghans** and U. H. E. Hansmann, *Numerical Comparison of Wang Landau Sampling and Parallel Tempering for Met-enkephalin*, *Int. J. Mod. Phys. C* **17**, 817 (2006).

More information can be found on [my ResearcherID profile](#), [my Google Scholar profile](#), [my ORCID profile](#), [my ResearchGate profile](#) or [my Impactstory profile](#).

#### Book Chapters

3. L. Delle Site, A. Agarwal, **C. Junghans**, and H. Wang, *Adaptive Resolution Simulation as a Grand Canonical Molecular Dynamics Scheme: Principles, Applications, Perspectives*, in: C. Clementi (Ed.), *Multiscale Methods in Molecular Biophysics*, Series in Computational Biophysics, CRC Press (2016), ISBN 1482225700.
2. E. Apol, R. Apostolov, H.J.C. Berendsen, A. van Buuren, P. Bjelkmar, R. van Drunen, A. Feenstra, S. Fritsch, G. Groenhof, **C. Junghans**, P. Kasson, P. Larsson, P. Meulenhoff, T. Murtola, S. Pall, S. Pronk, R. Schulz, M. Shirts, A. Sijbers, P. Tieleman, B. Hess, D. van der Spoel, and E. Lindahl, *Gromacs User Manual Version 4.6*, [gromacs.org](http://gromacs.org) (2013).
1. **C. Junghans**, M. Praprotnik and L. Delle Site, *Adaptive Resolution Schemes*, in: J. Grotendorst, N. Attig, S. Blügel and D. Marx (Eds.), *Multiscale Simulation Methods in Molecular Sciences*, *NIC Series Vol. 42*, Jülich (2009), 359.

#### Other Publications

6. *Balancing the load: Los Alamos researchers develop code to distribute computation more efficiently and across increasing numbers of supercomputer processors*. [ASCR Discovery Story](#), [HPCwire](#), [insideHPC](#) and LANL Science Brief (2018).
5. **C. Junghans**, A. K. Hüttel and U. Müller, *Gentoo Linux: Quelltexte und Rolling Releases*, [c't Magazin](#) **16**, 162 (2012).
4. S. Eidenbenz, K. Davis, A. Voter, H. Djidjev, L. Gurvits, **C. Junghans**, S. Mniszewski, D. Perez, N. Santhi and S. Thulasidasan, *Optimization Principles for Codesign applied to Molecular Dynamics: Design Space Exploration, Performance Prediction, and Optimization Strategies*, in: Proceedings of the DOE ASCR Exascale Research Conference, Portland, OR (2012), (LA-UR 12-20070).
3. T. Weidauer, **C. Junghans**, O. Pauluis, M. Pütz and J. Schumacher, *Shallow Moist Convection*, in: G. Münster, D. Wolf, M. Kremer (Eds.), *NIC Symposium 2010*, *IAS Series Vol. 3*, Jülich (2010), 373.



2. **C. Junghans**, M. Bachmann and W. Janke, *Phase Separation in Peptide Aggregation Processes - Multicanonical Study of a Mesoscopic Model*, in: U. H. E. Hansmann, J. Meinke, S. Mohanty and O. Zimmermann (Eds.), *From Computational Biophysics to Systems Biology 2007*, NIC Series Vol. 36, Jülich (2007), 169.
1. **C. Junghans** and U. H. E. Hansmann, *Cross-Check Methods in Protein Simulations*, in: U. H. E. Hansmann, J. Meinke, S. Mohanty and O. Zimmermann (Eds.), *From Computational Biophysics to Systems Biology 2006*, NIC Series Vol. 34, Jülich (2006), 157.